ISBN: 978-3-95450-263-9 ISSN: 2673-5520 doi: 10.18429/JACoW-MEDSI2025-WEP02

AS-BUILT FRONT ENDS FOR THE ADVANCED PHOTON SOURCE MBA UPGRADE*

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Abstract

The Advanced Photon Source (APS) upgrade from double-bend achromats (DBA) to multi-bend achromats (MBA) lattice is completed. All storage ring components and front ends were installed between April 2023 to April 2024 and fully commissioned. Some major changes have been made on front ends since preliminary design that was published in MEDSI'18 proceedings. The changes are a) Incorporated a Burn-Through-Mask (BTM) as the first fixed mask for all Insertion Device (ID) front ends; b) Removed clearing magnet from all front ends; c) Replaced beryllium window with diamond window in High Heat Load Front End (HHLFE). This paper presents the as-built version of HHLFE and Canted Front End (CFE).

INTRODUCE BURN THROUGH MASK

The APSU front ends functional requirement documents require all ID front ends to have sufficiently large inlet aperture (called front end acceptance) to accept all missteered beam given off by storage ring regardless of the storage ring beam dynamics. This acceptance is specified as ±1.67 mrad horizontal and ±0.67 mrad vertical. The required acceptance is the same for all ID front ends independent of whether it is HHLFE or CFE. This specification is very different from the original APS front end acceptance which depends on the type of front ends (single beam vs canted beam) and derived from storage ring beam dynamics. The First Fixed Mask (FM1) in original APS single beam front ends has the inlet aperture of 38 mm (H) × 26 mm (V) which is not large enough for the new APSU HHLFE. The FM1 which is located at 16.7 m needs to have an aperture at least 56 mm (H) × 22 mm (V) plus the half beam size. There is no space in front end to add an additional mask upstream of FM1, so the burn through mask (BTM) concept was developed. Before we can explain how BTM works, we need to explain how the storage ring beam position is monitored. Storage ring vacuum chamber and absorber are only able to withstand the heat load from dipole beam and not from undulator beam. To protect storage ring components from undulator beam strikes, there are two RF beam position monitors at each end of the straight section serving as Beam Position Limit Detector (BPLD). Based on APS operational experience, the BPLD is set at ±1.5 mm for horizontal and ±1.0 mm for vertical. These numbers already contain a 0.5 mm error margin. When electron beam position is out of that limit, the stored beam will be dumped to protect storage ring components. For front end BTM, when beam is within the BPLD limit, beam will just pass through the BTM aperture to reach FM2. If the BPLD is off or malfunctioning, the beam can burn through the front face of the BTM. There are air vent slots behind the mask face that cover the entire beam striking region defined by the maximum beam missteering defined by the storage ring aperture. The breach of the air vent will trigger the Storage Ring Valve (SRV) to close and dump the storage ring beam. The schematic of how the BTM works is shown in Fig. 1. Since the beam passing through the BTM can be fully captured by FM2, the FM1 no longer performs a function and is removed.

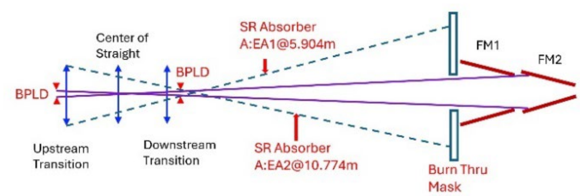

Figure 1: Schematic of missteering, BPLD and BTM.

HIGH HEAT LOAD FRONT END

The layout of the new HHLFE is shown in Fig. 2. The apertures of key components are shown in Table 1.

Table 1: HHLFE Aperture Table for APSU MBA

Components	Aperture H×V (mm)
Low Power Photon Shutter (LPPS)	130×40 shutter blade size
BTM	40.6×35.2 (inlet)/ 26×20.6 (out)
FM2	27.7×22 (inlet)/ 9×12 (outlet)
Lead Collimator	19.5×19.5 (optical) 26×26 (shielding)
GRID-XBPM	15.3×50 (inlet)/ 1.6×50 (outlet)
FM3	16×48 (inlet)/ 3.6×6 (outlet)
Photon Shutter	10×48 (inlet)/ 5×48 (outlet) at open
Safety Shutter	72×20 (optical/shielding)
Wall Collimator	27×17 (optical) 37×26 (shielding)
Exit Mask	12×48 (inlet)/ 2×1 (outlet)
Diamond window	5.5 dia. 0.2 mm thick
Exit Collimator	5×5 (optical/shielding)

* Work supported by the U.S. DOE Office of Science-Basic Energy Sciences, under Contract No. DE-AC02-06CH11357

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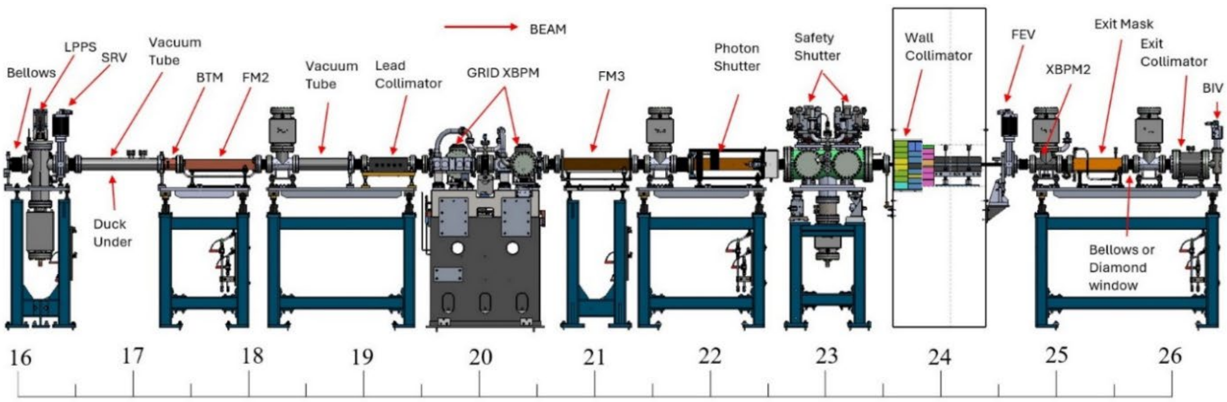

Figure 2: Layout of the as-built HHLFE for APS MBA upgrade.

HHLFE is designed for inline undulators. Most high heat load components were existing designs. The heat load limit is the combination of total power of 21 kW and peak power density of 590 kW/mrad^2 [1]. This heat load limit was used as the design requirement for the APSU HHLFE new components.

HHLFE starts with a Low Power Photon Shutter (LPPS) followed by a Storage Ring Valve (SRV) to isolate the front-end vacuum from storage ring. The LPPS can handle the power and power density from the dipole radiation, but not the ID beam. The LPPS and SRV table assembly is the same for HHLFE and CFE. With the LPPS and SRV closed, the storage ring can operate without the rest of the front end being functional. A spool piece replaced the previous FM1. This spool piece provides a space for duck under to access storage ring from the front-end side. The SRV will dump the storage ring beam if any vacuum failure of the BTM is detected.

The BTM design is shown in Fig. 3. It has two functions. It serves as the premask to block all dipole radiation at normal operation. It also serves as sacrificial device when BPLD malfunctions. The BTM allows the beam that is within the BPLD limit to pass through its aperture. If the BPLD is off or malfunctions, once the ID beam center is out of the green box and reaches the red line or beyond, the BTM can reach melting temperature. If the burn through occurs and reach the vent layer which is a 1 mm thick air pocket, the vacuum breach will be detected and trigger SRV to close which will dump storage ring beam.

FM2 is redesigned to be longer and have a slightly larger inlet aperture to accept the ID beam that is within the BPLD limit. FM2 provides the required aperture for the GRID-XBPM and protect the lead collimator aperture from synchrotron beam. Refer to reference [2] for the detailed description of the GRID-XBPM. The FM2 is box-cone design confining beam both horizontally and vertically and is made from solid GlidCop bar. The third fixed mask (FM3) protects all components downstream of it. The photon shutter and safety shutter are existing designs.

The exit mask was redesigned to remove the side port and all the XBPM2 detectors. The new exit mask is just a mask to provide fluorescence when beam hits, the detectors are now contained in the chamber upstream of the exit mask for ease of maintenance. A 0.2 mm thick diamond window of with 5.5 mm diameter clear aperture replaced all beryllium windows for high heat load handling purpose. APSU has a total of 19 HHLFEs and 12 of them windowless and 7 of them has a diamond window.

The space between the FM2 and collimator was set aside for a "clearing magnet" in preliminary design [3]. After a cost-benefit analysis and multiple internal and external reviews, APSU will not implement this component. A clearing magnet would use very strong permanent magnets that require special handling (similar to the magnets used in insertion devices) and would complicate assembly and maintenance in front end. Because the magnetic field must be checked regularly for any degradation, upkeep would also be very expensive. APSU will rely on storage-ring simulations to control electron trajectories rather than installing a clearing magnet.

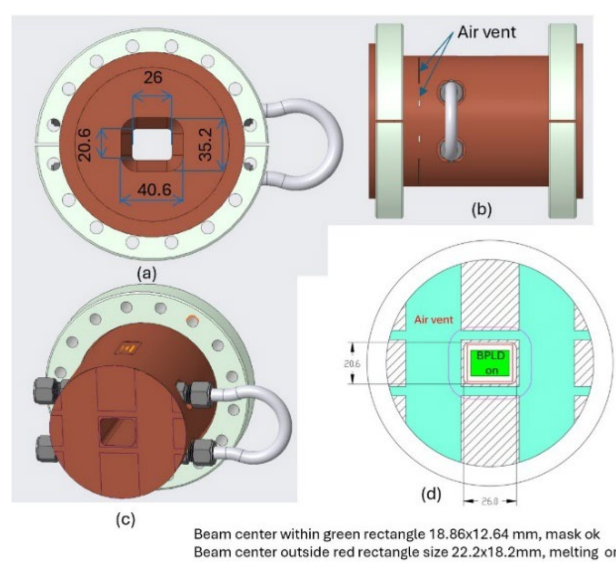

Figure 3: HHLFE BTM design, (a) inlet view, (b) side view, (c) section view of the air vent, (d) air vent details.

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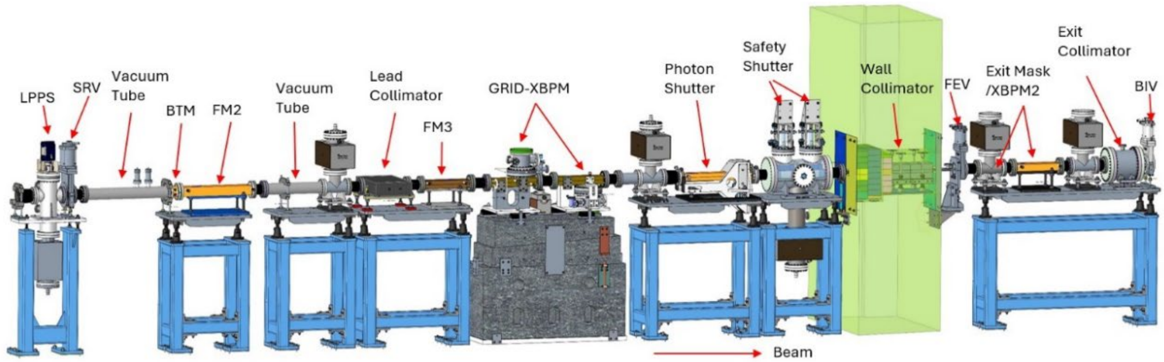

Figure 4: Layout of the as-built CFE for APS MBA upgrade.

CANTED FRONT END

CFE is designed for two canted beams with 1.0 mrad canting angle. Prior to APS upgrade, there were 7 existing CFEs. Most components in the existing CFEs were harvested. The high heat load components in the existing CFEs are designed to handle the heat load limit of 20 kW total power, which is 10 kW per beam and 281 kW/mrad^2 peak power density [4]. This heat load limit is sufficient for the APSU and is used as the design requirement for the APSU. The final design layout of the CFE is shown in Fig. 4 and is very similar to the layout of HHLFE. The first table is identical to that of the HHLFE. The FM1 is replaced by BTM. The horizontal aperture of BTM is larger to account for the 1.0 mrad beam separation. FM2 is an existing design to protect lead collimator. FM3 is a splitter mask to provide two separate beams for the GRID-XBPM. The principle for the GRID-XBPM in CUFE is the same as that in HHLFE. It is rotated 90º. CUFE GRID-XBPM has inclined scattering surface intercepting beam with vertical incidence and detectors mounted on the top and bottom. To prevent the cross talk of the two beams, a divider must be installed in the center of the GRID-XBPM. The divider is protected from beam strike by the FM3. The photon shutter and safety shutter are existing designs. The exit mask is redesigned to prevent beam cross talk. A new exit collimator design uses in-vacuum tungsten. The new design components for CFE are shown in Fig. 5.

Table 2: CFE Aperture Table for APSU MBA

Components	Aperture H×V (mm)
LPPS	130×40 (blade size)
BTM	59.6×30.6 (inlet)/ 45×16 (outlet)
FM2	46×17 (inlet)/ 26×5 (outlet)
Lead Collimator	40×16 (optical) 46×22 (shielding)
Splitter mask FM3	50×10 (inlet)/ dual 10×4 (outlet)
GRID-XBPM	50×10 (inlet)/ 50×1.5 (outlet)
Photon Shutter (PS)	50×10 (inlet)/ 50×5 (outlet) at open
Safety Shutter (SS)	50×16 (optical/shielding)
Wall Collimator	47.6×16.8 (optical) 56×26 (shielding)
Exit Mask (EM)	Dual 24×9 (inlet)/ 2×2 (outlet)
Exit Collimator	Dual 5×5 (optical/shielding)

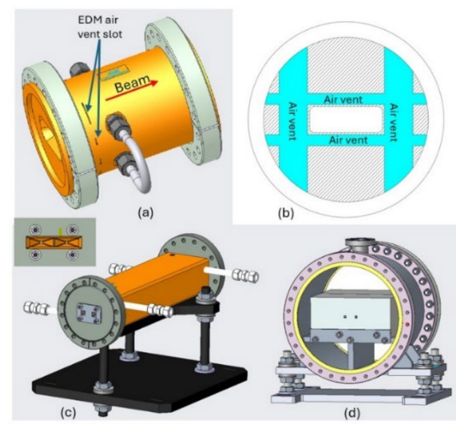

Figure 5: CFU components: a) BTM, b) BTM section view of air vent, c) exit mask, d) exit collimator.

CONCLUSION

A total of 19 HHLFEs and 16 CFEs were installed and fully operational. The traditional FM1 is replaced by BTM for all ID front ends. The clearing magnet in front end was not implemented.

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